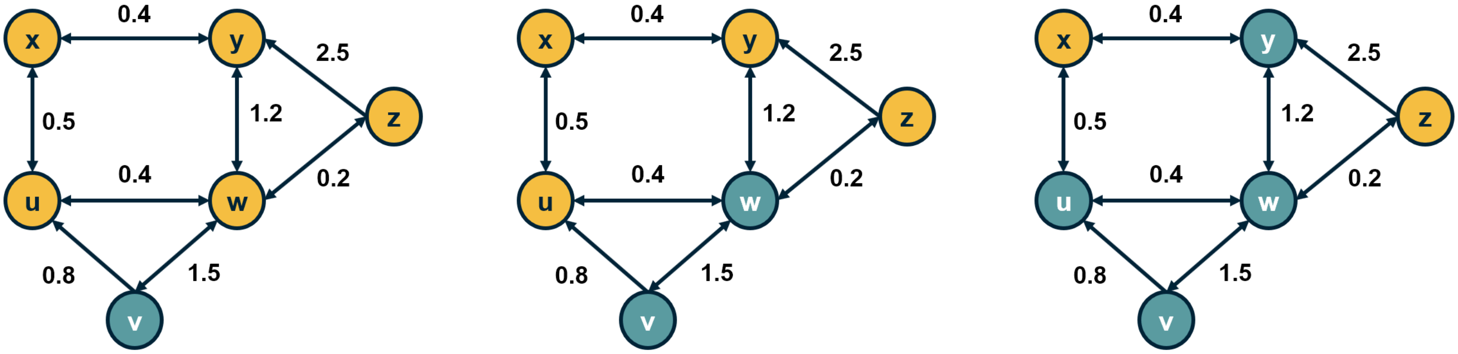
L10: Linear Threshold Model



Let us now see how we can model influence and social contagion with network models.

Consider a weighted *(and potentially directed)* network. The weight of the edge $w_{u,v}$ represents the strength of the relationship between nodes *u* and *v*. If the two nodes are not connected, we set $w_{u,v} = 0$.

The state of a node *v* can be either ***"inactive"*** $(s_v = 0)$ or ***"active"*** $(s_v = 1)$.

In the context of social influence, for example, an inactive node may not be exposed to a certain behavior *(e.g., smoking)* or it may be that it has been exposed but it has not adopted that behavior.

Initially, the only active nodes are the sources of the cascade.

Each node v has a threshold θ_v. The Linear Threshold model assumes that a node *v* becomes active if the cumulative input from active neighbors of *v* is **greater** than the threshold θ_v:

$s_v = 1 \text{ if } \sum_u s_u w_{u,v} > \theta_v$

Note that nodes can only switch from inactive to active once.

The Linear Threshold model is appropriate in diffusion phenomena when the ***"critical mass"*** theory of social influence applies.

An important question is: if we only activate a certain node v_0, which are the nodes that will eventually become active? These nodes define the "activation cascade" of v_0. Note that this cascade may cover the whole network, may include only v_0, or it may be somewhere in between. Of course, the cascade of v_0 includes nodes that are reachable from v_0.

A common simplification of the linear threshold is the **homogeneous case** in which all nodes have the same threshold θ. In the visualization, the threshold is set to $\theta = 1$ for all nodes. Note that the nodes *x* and *z* will not be part of the cascade.

There are also variations of the model in which two behaviors A and B are spreading at the same time, meaning that the state of a node can take three different values *(inactive, A and B).* In that case, the state of a node can switch between states A and B over time.

Another common variation is the ***"Asynchronous Linear Threshold"*** model. In that case, each edge has a certain delay. This means that different nodes can become active at different times.

The edge delays can affect the temporal order in which nodes join the activation cascade – but they cannot change the size of the cascade. We will review an application of this model on brain networks at the end of this lesson.

Note: Depending on the literature, the activation threshold model can be greater than or greater than or equal to - this can vary.

Food For Thought

Explain why the size of the cascade does not depend on edge delays.